

Design thinking for city dashboard development: Recommendations from a study of smart asset management in Sydney, Australia

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Abstract

The city dashboard has become an integral component of smart city asset management systems. It leverages data collected from multiple sources to monitor performance and enable evidence-based decision making. This article investigates the use of a design thinking framework to develop a functional and easy to understand city dashboard. The Smart Social Spaces project is used as a case study to illustrate how design thinking can be employed to develop an asset management dashboard, enabling efficient management of public space and infrastructure. The article profiles the unique collaboration between a local government, a multi-disciplinary team of university academics, and a street furniture designer and manufacturer, all located in Sydney, Australia. We unpack some of the design practice nuances that led this project to receive national awards and international recognition, and most importantly, created a user-friendly system to track and maintain public micro assets. We conclude with lessons learnt and recommendations for dashboard development through a design thinking process.

KEYWORDS

design thinking, city dashboards, smart asset management, collaboration, local government

Résumé

Le tableau de bord est devenu une composante intégrale des systèmes de gestion des actifs de la ville intelligente. Il exploite les données collectées à partir de multiples sources pour superviser les performances et permettre une prise de décision fondée sur des données exhaustives et à jour. Cet article analyse l'utilisation d'une approche design pour développer un tableau de bord urbain fonctionnel et facile à comprendre. Le projet Smart Social Spaces (Espaces Sociaux Intelligents) est utilisé comme cas pour illustrer comment l'approche design peut être employée pour

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concevoir un tableau de bord permettant une gestion efficace de l'espace public et des infrastructures. Plus précisément, ce texte décrit la collaboration unique entre un gouvernement local, une équipe pluridisciplinaire d'universitaires et un concepteur et fabricant de mobilier urbain, tous situés à Sydney, en Australie. Nous analysons certaines des nuances de la pratique du design qui ont permis à ce projet de recevoir des prix nationaux et une reconnaissance internationale, et surtout de créer un système facile à utiliser pour suivre et entretenir les petits équipements publics. Nous concluons par des recommandations sur la réalisation de tableaux de bord axés sur l'approche design.

MOTS CLÉS

pensée design, tableaux de bord urbains, gestion intelligente des actifs, collaboration, gouvernement local

Key messages

- Presenting city performance as a readable dataset that can be clearly communicated, while being visually appealing to a lay audience, is a challenge.
- An interdisciplinary design thinking process applied to dashboard design puts people both at the centre of and in the design process, as well as being at the centre of the outcome.
- The Smart Asset Management Dashboard served as a catalyst to change data infrastructure and integrate other asset data into a centralized platform for a Sydney-based local government council.

INTRODUCTION

In 2016, the Australian Government committed AUD\$50 million to a competitive grant program supporting projects that “apply smart technology, data-driven decision making and people-focused design to deliver economic, social and environmental benefits in metropolitan and regional urban centres” (Australian Government, Department of Industry, Innovation and Science, 2018, p. 2). The grant promoted collaborations between local government, private industry, and research institutions to deliver smart projects to improve liveability, productivity, and sustainability. Regardless of size or geographic location, councils can benefit from the use of smart technology that generates data to inform open space, urban design, and public infrastructure planning and decision making. A challenge for small local government areas is that they may not be able to afford the technical infrastructure needed to run a smart city management system without support from other orders of government (Rahmat et al., 2020).

In this article, we report on the design and development of a data-driven asset management system that integrates uncomplicated and aesthetically pleasing disparate urban data sources on a city dashboard. The project was executed over two years under the Australian government's Smart Cities and Suburbs Program, via a grant awarded to an interdisciplinary group of academics from the University of New South Wales Sydney and the University of Sydney, with Georges River Council as the local government partner. Street Furniture Australia was selected as the industry partner because of their 40+ years of designing and manufacturing durable furniture for the public realm. Rahmat et al. (2020) drew on a triple-helix model to frame this collaboration. Etzkowitz and Zhou (2017, p. 2) describe this as an “optimum methodology for entrepreneurship and innovation, moving research/knowledge in practice/use.”

Georges River Council, a government area with a 2022 population of 153,055, is situated in the southern suburbs of metropolitan Sydney (Figure 1), about 16 kilometres from the central business district (idcommunity, 2023). The council wanted to expand the use of public open spaces and amenities and improve its asset management system by adding a city dashboard. With two university research teams, an industry partner, and a substantial grant, the council had an opportunity to design, test, and install smart street furniture integrated with Internet of Things (IoT) sensor technology, and to pilot a city dashboard to manage these and other municipal assets.

We begin with a discussion of smart city dashboards. We then briefly discuss design thinking theory, its traditional application in industry practices, and its theoretical and practical relevance. This is followed by the Georges River Council case study and how a design thinking

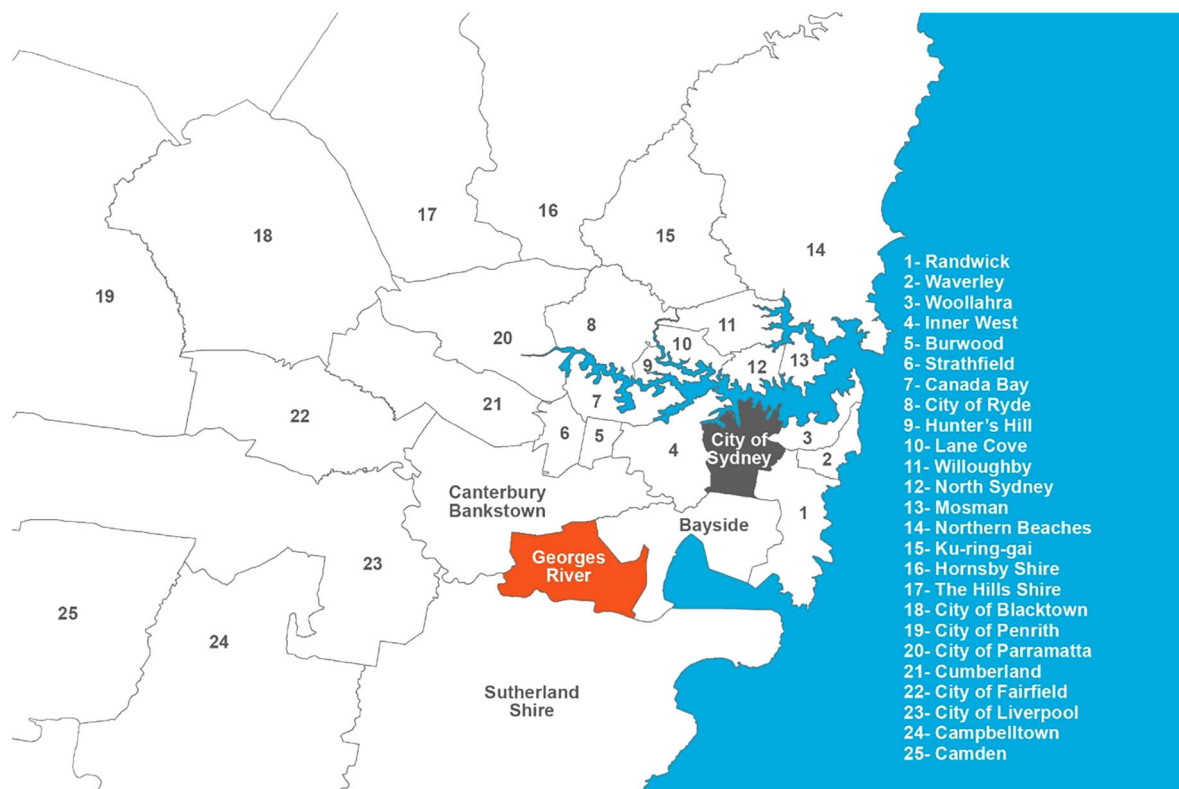


FIGURE 1 Georges River Council in the Sydney Metropolitan Area. Source: Authors (2019).

framework was used when creating the city dashboard. We conclude with lessons learned and recommendations for developing a design-oriented dashboard that constitutes a core component of any data-driven solution for smart cities.

SMART CITY DASHBOARDS

Local governments—urban, rural, and regional—are using data-driven technologies to make better, evidence-based decisions, in their aim to becoming smarter communities for their citizens (Rahmat et al., 2020). City-focused data can be found online on council websites, embedded in university research centre websites, or as stand-alone social media applications. City statistics offer citizens, analysts, policymakers, and local governments access to data about their immediate neighbourhood or surrounding suburbs. Data provides insight into city performance in areas such as transportation, infrastructure, energy, and environment, often in real time (Young et al., 2020). Public assets with sensors transmit data as part of the IoT, which enables municipal staff to monitor the assets, respond to immediate issues, and plan for future needs. One challenge accompanying the mountain of available data is unpacking it to make it meaningful and useful. Another is presenting that data in an accessible format that is easy to navigate, read, and understand for end-users with varying degrees of interest, skill, and expertise. One solution is the creation of a municipal or city dashboard.

Municipal dashboards are online visualizations of city data, often nested within the city website and featuring one or more webpages that present data specific to what is happening in the city in near real time. Sarikaya et al. (2018, p. 683) observe that “the dashboard concept has evolved from single-view reporting screens to include interactive interfaces with multiple views and purposes, including communication, learning and motivation, in addition to the classic notions of monitoring and decision support.” A dashboard may show weather conditions, traffic congestion, water quality, beach conditions, or presence (people) monitoring. For example, see the city dashboards for Dublin (Smart Dublin, 2016), Barcelona (Barcelona Digital City, n.d.), and Vancouver (City of Vancouver Open Data Portal, 2023). City dashboards are ubiquitous and Sadowski (2021, p. 1) believes dashboards have become “an essential part of the digital governance arsenal.” From Israel's Tel Aviv municipal dashboard offered in Hebrew, Arabic, and English, to Australia's Canterbury-Bankstown city dashboard offered in Arabic, Chinese, Greek, and English, the dashboard is becoming part of business as usual on many local government websites.

Dashboards are an integral part of smart city systems and services that leverage data collected from multiple sources such as sensors, mobile devices, social media, and official databases. Implemented correctly, dashboards can be the front-facing end of a data-driven system of

networks, enabling evidence-based decisions to be made to address urban issues. Dashboards are often viewed as data repositories to be managed, used, and interpreted by data analysts, engineers, or others with high data literacy. It is only recently that dashboards have been seen as visual tools that could offer data for purposes such as research (Sarikaya et al., 2018).

Dashboard development has traditionally been considered less of a design challenge and more of a technical one to be solved by experts. Sarikaya et al. (2018) argue that we need to stop thinking of dashboards as a unique method of organizing, presenting, and using data, but rather as an entity that covers a diverse set of practices meeting the needs of different people, purposes, contexts of use, and data environments. They highlight the need for a design framework that can accurately describe dashboard requirements—such as purpose, context, and users—in multiple applications, and claim that current models of dashboard development fail to capture the extent and complexity of this emerging space. Similarly, it has been acknowledged that there is a gap in the overall design process, specifically, taking account of users' expectations and skills (Young et al., 2020; Young & Kitchin, 2020). Young and Kitchin (2020, p. 5) argue that “community informatics and city dashboards in an urban informatics context should, therefore, focus on the evaluation of perceived usefulness and ease-of-use of new technology from multiple stakeholder viewpoints.” They recommend user-centred design for city dashboards to improve ease of use for three main user types. While the process of designing and developing the Dublin Dashboard has been documented (McArdle & Kitchin, 2016), a thorough discussion of whether and how user-centred practices have been adopted beyond Dublin is absent.

The design process adopted by dashboard developers and their clients is usually executed in private, where little is known about the methods used or how they reflect user-centred practices. The extent to which this leads to lack of public use or engagement with dashboard technologies is often only revealed later. As Sadowski (2021, p. 3) argues, accounts of “frustration and failure” are the reality of the adoption of new technologies that are not planned or crafted to present as a seamless system. Sadowski's ethnographic research into the development of two city dashboards in Australia critically examined some of the failures that occurred during the design process:

[F]unctional prototypes often stem from a mismatch of desire and capacity ... [T]hey have been told by consultants and companies that being data-driven is crucial for doing ‘smart’ governance. ... [H]owever, they do not necessarily have the resources, knowledge or ability to readily implement the technology. (Sadowski, 2021, p. 9)

Design aesthetics and visual appeal are at the crux of a dashboard's graphic interface. Kitchin et al. (2015, p. 10) note that visualization is “a key means of analysing and interpreting indicators and benchmarking data used to reveal the structure, pattern and trends of variables.” The interactivity features that allow users to filter, search, and layer data, and to easily navigate and control the display of data, are critical for making sense of the data and gaining insights beyond statistics (Haeusler et al., 2017). Interactive and responsive design provides greater flexibility to modify many attributes, such as the construction and composition of views, drill-down options, the level of detail available, search functions, and comparison functions. Flexibility also offers adaptability to different users, situations, and display devices or multi-display environments (Sarikaya et al., 2018).

Like most integrated systems, however, city dashboard design does not come without challenges such as:

- Varying formats of data representation disrupting what should be seamless integration (Batty, 2015).
- Integrating diverse data sets (demographics) and types of data (geospatial, sensor) to make them meaningful and accessible to a wide range of users.
- Adapting platforms for use on multiple devices such as mobile phones or tablets.
- Difficulty in establishing a universal iconography to represent qualitative variables (e.g., community awareness, engagement, and trust) (Sarikaya et al., 2018).

Cautions have been raised about the use and functionality of municipal dashboards, as well as about over-reliance on them. These concerns relate to the nature of the city data and its performance, which is complex, layered, and not always easily collected, measured, or synthesized (Batty, 2015; Kitchin et al., 2015). Presenting city performance as a readable dataset that can be clearly communicated, while being visually appealing to a lay audience, is a challenge—both for city councils wanting to communicate with their citizens and for the software and graphic designers whose job it is to make meaning from data so others can readily interpret it.

Mattern (2015) argues that a challenge for dashboard designers is the amount and accessibility of data about our cities. She argues that “urban dashboards perpetuate the fetishization of data as a ‘monetizable’ resource and a positivist epistemological unit—and they run the risk of framing the city as a mere aggregate of variables that can be measured and optimized. ... [M]uch of what we perceive on our urban dashboards is sanitized, decontextualized, and necessarily partial” (Mattern, 2015). In the case of the Open Michigan Dashboard, benchmarking and metrics for the state's progress categories (e.g., education, health, wellness) were available for public consumption. However, the methodology for how the data were obtained was vague, leaving much to be desired in terms of the reliability and validity of the dashboard information (Mattern, 2015). In Australia, many city dashboards are realized through university research and local government collaborations for council uplift and in response to a local smart strategy. For example, see the City Futures Research Centre at the University of New South Wales (University of New South Wales,

2023). Other dashboards appear to be a single page with minimal information. As Sadowski (2021, n.p.) critically observed, some dashboards are not maintained and never evolve, becoming “obdurate relic[s] of old decisions, old technologies and old bureaucracies.”

DESIGN THINKING THEORY AND MODELS OF PRACTICE

Design thinking as a protocol was developed in the early 1990s by David Kelley, founder of the global design and innovation firm IDEO. The premise of design thinking is that it is a collaborative process involving people who may never have thought of themselves as designers (Brown, 2009; Rawsthorn, 2013). This quality makes it unique as a design approach. The core principle is that people are both at the centre of and in the design *process*, as well as being at the centre of the *outcome*. In a design thinking environment, creativity is highly valued and encouraged among those who might otherwise see themselves as the non-creatives in the group.

In the early stages of a project, experimentation is encouraged. There is an acceptance that “there is nothing wrong with failure, as long as it happens early and becomes a source of learning” (Brown, 2009, p. 230). Despite the growing popularity of design thinking and its ability to deal with problems in information technology, business, education, and medicine (Dorst, 2011; Valentine et al., 2017; Linton & Kinton, 2019; Sohaib et al., 2019), its potential to assist in developing data-driven systems and products in a smart city context has remained largely unexplored. In addressing this gap, we present examples via a case study, demonstrating the potential of using interdisciplinary design thinking processes to reframe how we use data analytics and visualization in the creation of city dashboards.

Design thinking has been adopted across many sectors, notably in business and service design. It is an iterative process, broad in scope, and often diverges from the original topic (Brown, 2009). The process involves rapid ideation where participants observe and collaborate in exploring design solutions through fast learning from quick prototypes and testing (Lockwood, 2009). This approach to business development and innovation emerged as a response to the rising complexity of organizational cultures, customer-service oriented businesses, and the increased value of creativity and collaboration as skill sets in the workplace (Kolko, 2015).

Fundamentally, design thinking is a flexible process (Miller, 2017; Hasso Plattner Institute of Design, 2018). Carlgren et al. (2016) identify five common practices in design thinking which lead to innovation or problem resolution. Design thinking needs to be **user focused**, meaning there is a deep recognition of the need for user groups and user involvement. Through **problem framing**, there is an expanded solution space through divergent thinking, where multiple scenarios are explored. This entails repeated questioning and reformulation of the initial problem. An exercise of **rapid prototyping** distills the scenario imagined into a low-resolution representation or mock-up of ideas or solutions. This leads to **experimentation** with various iterations of the prototype, whereby multiple solutions emerge. What brings this process together are **diverse teams** with different skill sets and personalities (Carlgren et al., 2016). The proposed framework for developing city dashboards, presented later, is informed by these five components of design thinking.

A design thinking approach for innovative problem solving or creative service solutions has not reached all sectors. For example, smart cities were initially defined as a technology-intensive approach to improve the ways city services and infrastructure operate (Barrionuevo et al., 2012; Chen, 2010). Sadowski and Maalsen (2020) believe that initiatives driving smart cities are often *ad hoc*, where “existing stuff is upgraded and replaced, here and there, based on what resources are available, what is achievable, and what opportunities arise.” Stimmel (2015) argues that design thinking has not been consciously applied in the smart city space. It has only been recently that people have been explicitly included as a component of smart cities. Albino et al. (2015, 5) report that “the smart city concept is no longer limited to the diffusion of ICT [information and communications technology], but it looks at people and community needs” and in general, overall human needs rather than mere technical specifications (Stimmel, 2015).

Design thinking through observation, information gathering, ideation, and prototype development offers a way to actively involve end-users in the development of smart city innovations or projects such as city-focused apps (Tomitsch, 2018). It can also offer opportunities to match people's needs with technological feasibility, resulting in viable business strategies that can achieve both customer value and market opportunity (Brown, 2008). Design thinking can be used to involve citizens in the conceptual stages of developing smart city services and products. These are also known as people-centric intelligent services, a new trend that reflects strategies and technologies derived from citizens' requests and needs (Xu & Geng, 2019). It also requires transparency of prototyping to all stakeholders in the design process, meaning that prototypes are open to discussion and dialogue by both the design team and clients. Design thinking can be an essential tool for simplifying and humanizing the design process (Kolko, 2015), thus mitigating some of the technological complexities of products, services, and processes in smart city solutions.

Embedding design thinking in a city dashboard development: A framework

Design thinking can aid in finding a balance between the complexity and usability of dashboards. This balance is critical, as excessive features or feedback might negatively affect decision making (Yigitbasoglu & Velcu, 2012). Figure 2 is a schematic diagram of the application of design

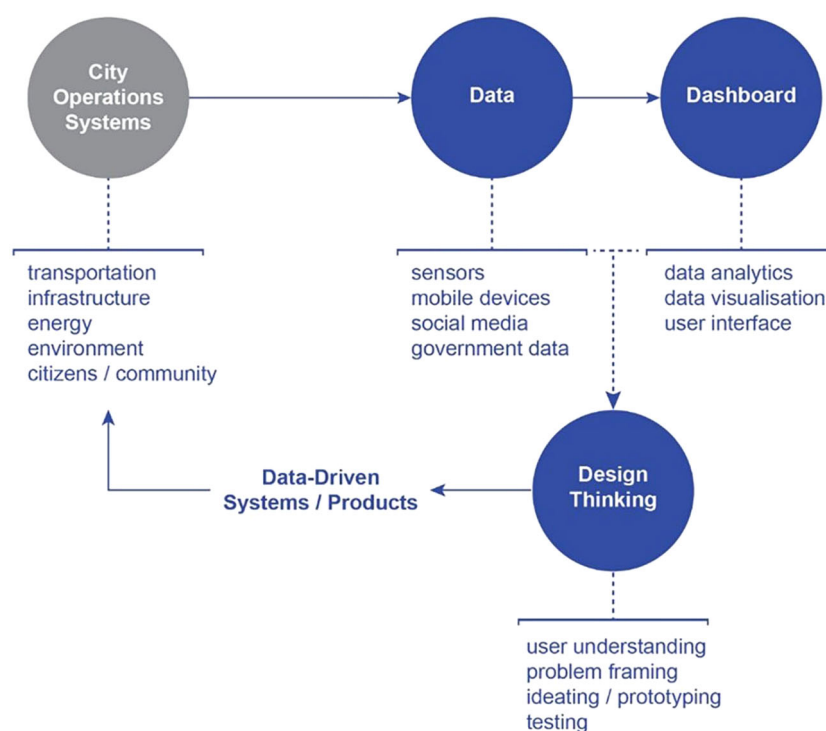


FIGURE 2 Schematic diagram of the application of design thinking in the development of a city dashboard. Source: Authors (2020).

thinking in the development of a city dashboard. It demonstrates a typical city planning challenge of improving urban operations in areas such as transportation, infrastructure, energy, environment, and community needs.

The first stage of the design thinking process set out by Carlgren et al. (2016) is *focusing on the user* and understanding the human needs involved. Young et al. (2020) highlight the benefits of interviews with user groups to determine a range of user needs and their ability to engage with smart city technologies. This helps to avoid inefficient design that could result in a dashboard with limited functionality that is too broad or too narrow, serving a limited audience for a singular purpose. They contend that designers and developers should take account of users' expectations and skills and conduct a user evaluation analysis to inform a user-centred approach to city dashboard design and creation. Assessing the effectiveness of presenting information to all potential users, regardless of ability and knowledge, is vital to ensure the dashboard is user-friendly.

Understanding the human factors involved allows the creators to reframe the problem in human-centred ways (Dam & Siang, 2018). Site-based research and observations help generate place-based knowledge to inform the next stage, *problem framing*. This is especially important when developing city dashboards that involve complexity and multiple data streams.

In the visualization and *prototyping* stage, there is a key role for design thinking in developing the dashboard aesthetic as a communication tool. This can ensure that, beyond the needs of staff and council, community users can understand rich datasets and engage with their city in a new online environment (Moere & Purchase, 2011). Applying design thinking to the visual features of the dashboard is critical to enable users to make sense of the data and generate insights. To this end, a dashboard developer becomes engaged in a complex process of design to address three requirements: (1) utility—its functionality, usability, and usefulness; (2) soundness—its reliability and robustness; and (3) attractiveness—its aesthetic. Like other design-related fields, communicating data graphically and through a technology interface requires a balance between the product being useful, rich in content, and visually appealing (Moere & Purchase, 2011).

Experimentation is a distinguishing feature of a design thinking approach, entailing an iterative approach during prototype refinement. New solutions and services delivered through dashboards can be explored through prototypes in a non-linear and exploratory process. As Kolko (2015) notes, the iterative nature of the design process recognizes that it is rare to get things right the first time. In a design process, failure is leveraged as learning and viewed as part of the cost of innovation. In developing city dashboards, the cost is often a result of trial and error with feedback received from end-users. Through failure, though, it is possible to improve the functionality and visual quality of the dashboard before it is fully implemented. Feedback has the potential to result in new ways of presenting data and the addition of new features that complement existing ones. A design thinking approach starts small, enables fast learning, and scales up rapidly. Prototyping and experimenting enable the designers and developers to manage the functional, technical, and operational complexity of products, services, and processes (Kolko, 2015; Lockwood, 2009), while ensuring the result is user-friendly for a range of dashboard stakeholders.

CASE STUDY: SMART SOCIAL SPACES

This case study provides an example of how design thinking was applied when developing a data-driven asset management system and city dashboard for a local council. The system aims to address a common city operating challenge: the increasing pressure on public infrastructure. While local councils must provide a level of amenity in public spaces, they are also tasked with managing urban assets installed in these places. The Smart Social Spaces project sought to address the local council's need for an effective, affordable, and relevant system that would provide hard evidence from IoT devices to inform open space, urban design, and public infrastructure planning and decision making. As a recently amalgamated local government area, Georges River Council wanted to be more efficient as an organization while delivering higher quality and better integrated services and facilities for its residents. Council saw an opportunity to partner on this project to provide better public spaces by using smart infrastructure to create a cost-effective and technologically manageable asset management system.

IoT devices such as smartphones, sensors, and cameras can measure microclimates, utility use, and social behaviour, providing basic information about local conditions. When used as an interconnected framework, IoT systems provide connectivity between sensors and the internet, which enables end-users to interact with smart objects (Mineraud et al., 2016). When analyzed as a system of devices, it can have a significant impact by decreasing the operating costs of urban services. IoT devices can monitor air quality and the structural health of public assets, and aid in waste management, noise assessment, and traffic flow. They can monitor energy resource distribution and consumption, as well as parking, lighting, and water supply (Difallah et al., 2013; Lee et al., 2013; Zanella et al., 2014). Tanda and De Marco (2018) propose that one of the main factors driving the implementation of IoT-enabled smart city services is the potential economic savings for a city through the management of a variety of urban assets from within one system. The smart system based on IoT technology developed for the Smart Social Spaces project is an example of IoT-enabled smart city services (see Steinmetz et al., 2021).

Incorporating IoT sensor data in a dashboard for smart asset management

Sensors were embedded in new street furniture assets to measure a range of real time events and environmental information. For example, smart bin sensors detect bin fill levels to facilitate efficient scheduling for emptying. They also have the capacity to read internal bin temperatures and notify Council Facility Management of potential fire hazards. The Healthy Living Hardware pole, a new street furniture typology developed and tested for this project, includes an array of sensors that can detect the presence of people within a limited perimeter. Park seating arrangements detect if the seating is being used, a meter in the electric barbecue facilities measures power consumption, and another meter measures the sound level in the public space. Data are collected from IoT sensors, transferred to, and stored in cloud servers via a Wi-Fi or a 3 G/4 G network, and then processed and visualized on the dashboard. Figure 3 shows the complexity of the sensor matrix behind the project's data-driven asset management system, which is presented on the front-facing dashboard.

The Smart Asset Management Dashboard development process and system design

The Smart Asset Management (SAM) Dashboard was developed as a web-based interface to present filtered, aggregated data acquired via the platforms associated with the vendors of each sensor used. Machine to machine translations of the raw data through a vendor Application Programming Interface and website scraping were collated and filtered to be displayed on the SAM Dashboard, bringing together multiple sources of data in a centralized platform.

Design thinking was central to the development of the SAM Dashboard through a non-linear process in which all stages were not weighted equally. First, user understanding and problem framing were incorporated to identify the purpose of the dashboard according to primary users. Next was visualization, prototyping, and testing the dashboard interface. The chief methods deployed during the process included co-design workshops and design charrettes with primary end-users, as well as an assessment of case studies on other city dashboards and their interfaces.

User understanding and problem framing for the SAM Dashboard

The primary end-users of the SAM Dashboard were city maintenance crews, environmental and infrastructure services, park crews, asset managers, and the design team. Members of the public were considered as possible future users of the dashboard so there was no consultation with them at this stage of the project. Through multiple co-design workshops, the SAM Dashboard design team consulted with council staff and the product development team of the street furniture designer and manufacturer to gain insight into their respective needs. The expectations of the co-design workshops were to capture the vision, values, ideas, and interdisciplinary expertise of the team. Different council "personas" were considered through a discussion of different roles and experience in council, levels of seniority, frequency of use, and why those groups needed a dashboard.

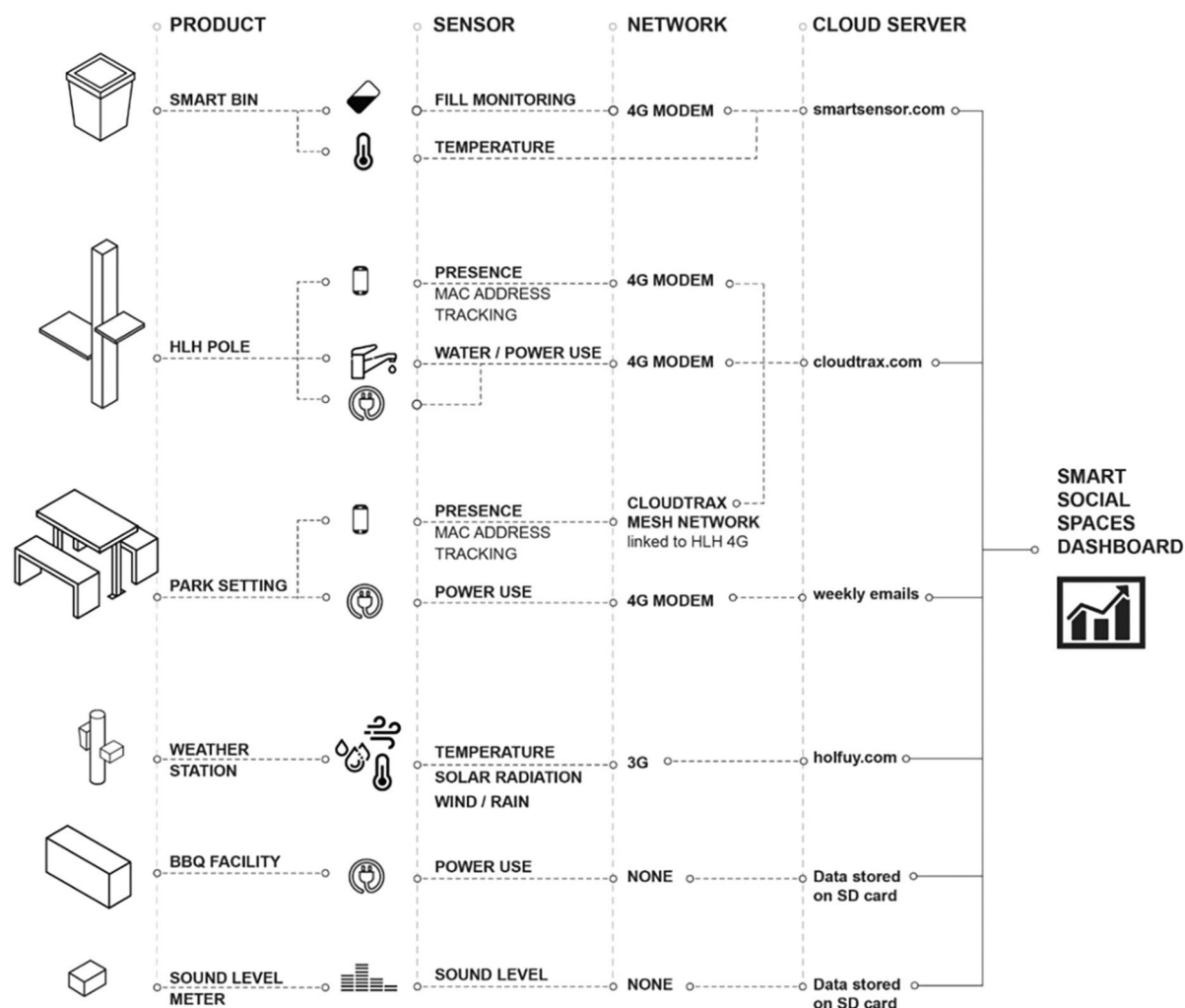


FIGURE 3 The sensor matrix and data-driven asset management system behind the project's dashboard. Source: Authors (2020).

The SAM Dashboard that was developed was intended to serve as an information hub for various audiences, including: (1) local council to improve asset management; (2) the design team, manufacturer, and researchers to assist with product development and service innovation; and (3) the wider community to offer local information such as microclimate data. Further enhancements of the community dashboard may provide the status of Wi-Fi and power, as well as a booking system for barbeques and picnic tables. Some new requirements arose, and designs were changed during the co-design workshops. For example, while the original purpose of the dashboard was to enable asset management, a secondary purpose was to support product development and service innovation by the industry partner.

A key feature of the SAM Dashboard is its system of monitoring and evaluating the use of public spaces and the effectiveness of the micro-infrastructure assets used within them (e.g., street furniture, bins, and other structures). The SAM Dashboard offers a digital twin of physical assets to monitor their use and manage maintenance. The digital model of the urban assets is fed with both historical and real time sensor data. By providing multiple sources of data, the dashboard can be used simultaneously for waste management, noise monitoring, environmental assessment, monitoring power and water usage, and presence monitoring (i.e., determining the number of people visiting a site and the duration of those visits).

The SAM Dashboard aims to improve cost efficiencies for council by supporting the operational maintenance of its micro-infrastructure assets. It helps staff evaluate the near real time status of assets by describing their current and past state through a data feed from the onboard sensors through the cloud to the dashboard.

Product performance data and environmental measurements obtained from actual in-situ use provided new understanding of how an IoT-enabled prototype such as a smart bin or bench is used. Through this, the SAM Dashboard offered the design team a means to gain high-fidelity, real time remote feedback during the testing of a new product or service. This function can enhance the quality of feedback received during the testing period or when commissioning a newly installed product. An advantage is that sensors provide an authentic reading of user interaction and feedback and are not clouded using a simulation or lab setting with an observed group of users.

Visualizing, prototyping, and testing the SAM Dashboard

The design team generated multiple ideas for the front-facing interface of the SAM Dashboard. Free divergent thinking was practiced, expanding the solution space and exploring different ideas for data visualization. The ideation process was informed by an assessment and case study of the design features of other city dashboards such as the London Dashboard (2023), Dublin Dashboard (Smart Dublin, 2016), Melbourne Dashboard (2017), Edmonton Dashboard (City of Edmonton, n.d.), as well as dashboards with a broader scope such as a City Health Dashboard (2023) and a Performance Dashboard (Australian Government, n.d.).

The visual prototypes were presented to council and the street furniture designer and manufacturer for pilot testing. Several design options and scenarios were explored with these end-users during intensive, hands-on design charrettes. This included the layout (i.e., presenting data as map view or an asset view), the balance of text and images, colour selection, the use of icons, and the preferences for presenting information on devices with small or large screens (i.e., desktops, laptops, tablets, or smartphones). Multiple sessions over time were held to ensure the feedback loop was in place before finalizing the design considerations. Given the wide audience of the dashboard, including people with lower levels of data literacy, data were represented in easy-to-read bar charts and pie charts.

Another important design consideration was to avoid cluttering the user interface. A key issue centred around how to integrate asset data with its location on a map and how to present data in a way that ensured a balance between utility, soundness, and attractiveness—the main requirements of information visualization recommended by Moere and Purchase (2011). During testing, interdisciplinary collaboration among stakeholders was pursued to position data visualization in context for both council and the street furniture designer and manufacturer. This ensured each group understood the full extent of the dashboard's use and implications for asset management, as well as for product development and service innovation.

The experimentation and iteration phase proved to be an important stage of design thinking, during which the dashboard's user interface was refined based on the critique and feedback of a range of end-users. This was particularly effective for improving data presentation and enabling a better understanding among various end-users such as the park manager, garbage truck driver, community planner, and street furniture designer who would be monitoring the dashboard offsite.

The resulting interface of the SAM Dashboard

Few (2006) suggests that interface design involves three steps: see the big picture, narrow in on specific critical features, and link to supporting details. The SAM Dashboard interface achieves this by providing a quick overview of the current state of numerous public assets in one place. Once an asset is identified as needing attention, additional details are available (including historical data) by clicking on each data point and specifying a time frame. The SAM Dashboard can also be configured to initiate an automated response in case of a power outage by sending emails alerting asset managers to act.

A key feature of the SAM Dashboard's interface is including both a user-friendly map view (Figure 4) and an asset view that presents a selection of tiles that lead to data on different types of street furniture (Figure 5). The tiled view provides key metrics and simple charts associated with an asset, along with the opportunity to zoom in for more detail. The preferred view is selected through a guide bar on the left side of the dashboard, where users can also explore physical sites or sensor status.

RECOMMENDATIONS FOR DASHBOARD DESIGN: LESSONS LEARNED FROM SAM

Five key recommendations for city dashboard design emerged from the use of design thinking to develop a Smart Asset Management Dashboard for Georges River Council in Sydney, Australia. The critical insights from the development of the SAM Dashboard by an interdisciplinary team are:

1. Engage with different types of end-users in the dashboard development process to account for their specific needs when searching for information.
2. Explore the dashboard's purpose and function over several iterations to accommodate the evolving understanding of different end-users' needs.
3. Provide in-house training to develop expertise among the end-users who need to manage and maintain the dashboard, digital infrastructure, and IoT sensors.
4. Undertake periodic evaluations to adapt to the changing needs and data literacy of the city staff and council members who use the dashboard.
5. Use design thinking to create a better dashboard interface than would have been possible with a technology-focused approach.

The main purpose of a city dashboard is to monitor current conditions. However, running a dashboard for a data-driven system that relies heavily on IoT sensors poses challenges and risks. Providing a stream of useful data relies on the stability of the IoT system and maintaining



FIGURE 4 SAM Dashboard interface: Map view. Source: Authors (2020).

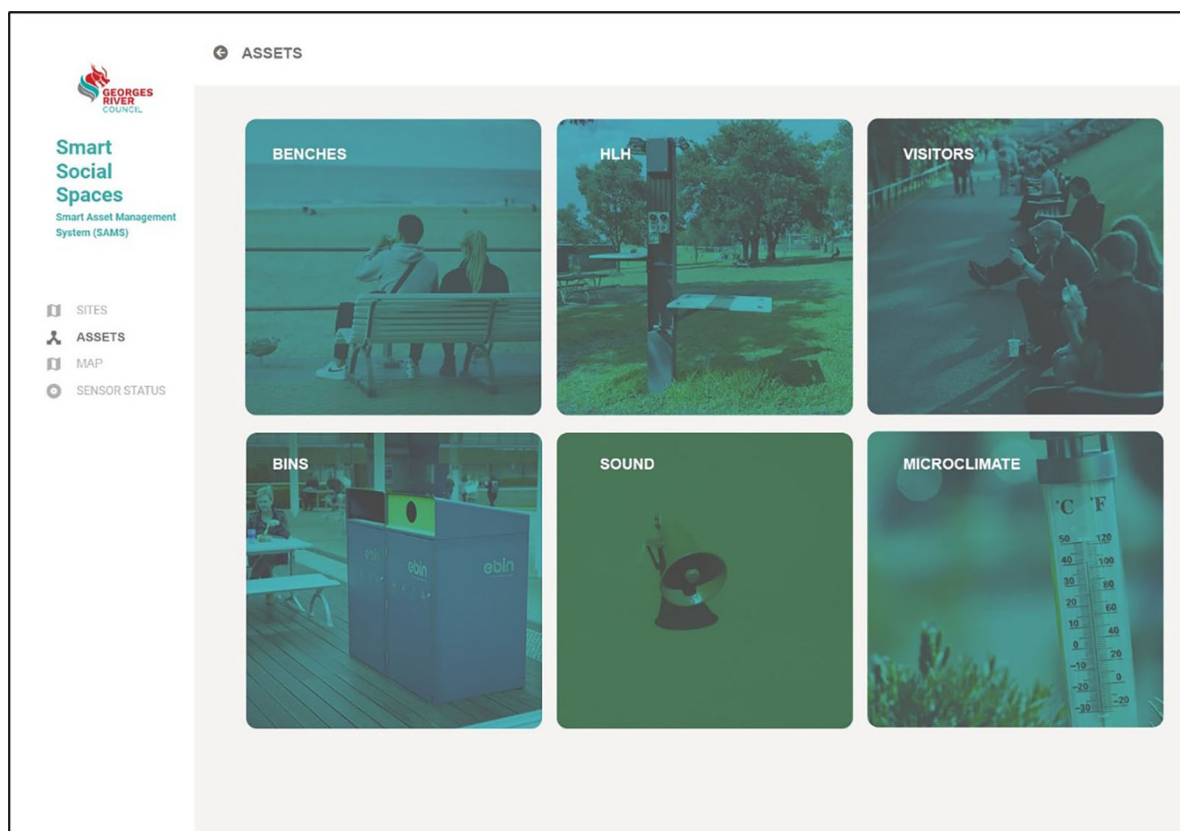


FIGURE 5 SAM Dashboard interface: Asset view. Source: Authors (2020).

operational sensors was an ongoing challenge during the project. Power outages, Wi-Fi dropouts, and vandalism resulted in a loss of data. Ongoing maintenance is essential to ensure sensors and networks are operating so they feed data into the dashboard. This was acknowledged by Mineraud et al. (2016) who noted that fault management is essential when using IoT devices that produce low-level data and that are often unreliable, incomplete, disordered, and can go missing.

The purpose of a dashboard is established at the outset and refined through iterative design thinking. Yet its functionality, user base, visual aesthetics, and data output should be re-evaluated periodically to ensure it responds to council requirements and remains relevant to the changing needs of the community. Thus, the dashboard and underlying data-driven system should be developed as a dynamic tool with opportunities to serve unknown future users and purposes. In the case of the SAM Dashboard, later in the project, council became interested in using the dashboard for purposes beyond asset management. They wanted data that could be used as evidence for an urban master planning project, which entailed developing new techniques of analysis and customized filtering.

Due to the digital nature of dashboards, they must be able to adapt to new information needs, changing user expectations, and innovative design trends. In the case of the SAM Dashboard, as additional smart infrastructure was added to the public space, the dashboard evolved. The refinement of any city dashboard, including its purpose and function, requires revisiting the context in which data will be disseminated and how and by whom they will be used. This periodic revision can make a difference “between numbers that just sit there on the screen and those that enlighten and inspire action” (Few 2007, p. 2). The right context enables end-users to compare, interpret, evaluate, and synthesize information.

A city dashboard fed by IoT sensors needs to be modified through an iterative process to ensure the collected data are accurate and reliable. The SAM Dashboard was created to understand when people were using or interacting with the street furniture. Media access control (MAC) address tracking via Wi-Fi was used to count footfalls as an indicator of the number of people near furniture assets. However, this is only a proxy, as individuals may be carrying multiple devices or none, and while they may be near an asset, they may not be using it. To complement this data, we used vibration sensors in park seating arrangements. Nevertheless, detecting the presence of people using or interacting with other furniture assets was not possible through this technique. By identifying this challenge in the testing stage, we came up with the solution of cross-checking data feeds and making sense of them in relation to each other. However, a clear display of this data also presented a challenge for the dashboard design.

The case study of the SAM Dashboard demonstrates that in the process of developing a smart city dashboard, multiple voices need to be heard and all parties should be involved in sharing and refining their expectations for the end-product. It is critical to understand how different users such as council managers, project officers, maintenance staff, and the public will interact with data. Shifting to a new system of technology requires additional work, resources, and adaptations for what might be an already overloaded council. As such, it was important to demonstrate to various end-users from the outset of the project the potential benefits and tangible outcomes of using a dashboard.

A dashboard that is linked to IoT-enabled devices introduces a new layer of digital infrastructure that requires significant extra resources, as well as new roles and responsibilities that often do not exist or fall between existing specialist roles. For example, a council-run IoT street furniture asset may be managed by the public works or park maintenance teams, while the sensors and data management may be handled by the IT department. In the case of the SAM Dashboard, to facilitate the utilization of the dashboard by city staff, an instructional video was produced to introduce the main features of the dashboard to support a seamless transition for its use.

The management of a digital twin of physical assets that includes the data from sensor networks and analytics requires new cross-disciplinary skills and responsibilities. Moreover, expertise and training to manage and deliver IoT projects also needs to be developed. For asset managers, it requires developing new expertise to maintain and manage IoT-equipped products.

The Smart Social Spaces case study illustrates how a city dashboard designed to support data-driven products can provide a flexible and smart infrastructure management system for a local council. The SAM Dashboard served as a catalyst to change council's data infrastructure and to integrate other asset data, including stored data, into a centralized platform to allow for better access, use, and sharing. The project reflects and promotes a progressive approach towards an integrated and functional use of data in urban governance, where making informed and critical decisions about urban infrastructure and city management is facilitated.

DESIGN THINKING AND CITY DASHBOARDS: WHERE TO NEXT?

A design thinking framework can contribute to the development of dashboards that have a clear purpose and fit the context where data is being used. Kitchin et al. (2015, p. 24), “advocate for indicator and dashboard projects that openly recognize their situatedness as the product of a complex data assemblage.” This means positioning the data in context, as well as understanding its use and implications. The approach here enabled the dashboard developer to go beyond real time data and provide meaningful information related to challenges and problems in Georges River Council. From the simple presentation of data to offering practical insights, an effective city dashboard can identify and analyze trends in city operations and in the performance of infrastructure and public assets.

When developing data-driven systems and products that enable evidence-based decision making and problem solving, it is important to start small, learn fast, and scale up (Lockwood, 2009). This enables developers and designers to manage the functional, technical, and operational complexity of products, services, and processes in smart cities. Solutions will entail integrating multiple databases with various forms, volumes,

and velocities. It is also crucial that all elements of a smart city data system can talk to each other. An interoperable, open source, and non-proprietary system allows future data reuse and the publishing of some of data streams to third parties. It also links to existing data repositories of council to facilitate scaling up and sustaining data-driven practices. Access to open data and real time information can also create communities of informed citizens and reveal opportunities for data-driven decision making and transparency in local governance.

Smart city systems, data collection, and analytics are not a substitute for detailed knowledge of an area and its people. City officials remain at the forefront of being able to interact with community members, which is a privileged position that cannot be replaced by big data analytics and automation. Sole reliance on a bank of computer screens and dashboards is doomed to fail. Councils and their staff need to be out and about observing people and places. As noted by Mattern (2017), urban intelligence is more than information processing, it involves site-based experience, observation, and sensory engagement to produce “new models for thinking about cities that do not compute.” Critical perspective is needed to transform urban data into meaningful spatial information and translate it into place-based knowledge (Mattern, 2017). This can complement site-based techniques for understanding and monitoring local conditions in cities.

CONCLUSION

As cities evolve and grapple with managing increasing volumes of data, the need for developing an effective dashboard as part of a data infrastructure system is widely recognized. Smart city dashboards are more than data repositories accessible to a limited audience. Design thinking was explored as an iterative problem-solving activity that integrates human needs with technology to develop data-driven solutions for asset management by using dashboards. The design thinking approach used to develop the SAM Dashboard focused on understanding the needs of a range of users, problem framing for the user groups, prototyping dashboard models, and iteratively testing them.

The case study of the Smart Social Spaces project demonstrates that asset management innovation and the use of smart analytics is achievable through the design and development of a dashboard that integrates multiple and disparate urban data sources. It was at times challenging to collect, measure, and synthesize the data into a readable dataset. In the end, local authorities were able to benefit from a smart public asset management system, attain value from limited resources, and provide a better quality of life for their citizens. The collaboration of university, local government, and industry personnel harnessed creativity and talent across disciplines and between the public and private sectors. Smart cities require this type of interdisciplinary knowledge and collaboration to address the challenges facing contemporary cities.

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REFERENCES

- .idcommunity. (2023). *Georges River Council: Community profile*. <https://profile.id.com.au/georges-river>
- Albino, V., Berardi, U., & Dangelico, R. M. (2015). Smart cities: Definitions, dimensions, performance, and initiatives. *Journal of Urban Technology*, 22(1), 3–21.
- Australian Government. (n.d.). *Performance Dashboard*. <https://performancedashboard.d61.io/aus>
- Australian Government, Department of Industry, Innovation and Science. (2018). *Department of Infrastructure and Regional Development, Smart Cities and Suburbs Program Round Two, Program Guidelines Smart Cities and Suburbs Program*. <https://business.gov.au/Grants-and-Programs/Smart-Cities-and-Suburbs-Program>
- Barcelona Digital City. (n.d.). *Municipal Management Dashboard*. <https://ajuntament.barcelona.cat/digital/en/digital-transformation/city-data-commons/municipal-management-dashboard>
- Barrionuevo, J. M., Berrone, P., & Ricart, J. E. (2012). Smart cities, sustainable progress. *IESE Insight*, 14, 50–57.
- Batty, M. (2015). A perspective on city dashboards. *Regional Studies, Regional Science*, 2(1), 29–32.
- Brown, T. (2008). *Design thinking*. <https://hbr.org/2008/06/design-thinking>
- Brown, T. (2009). *Change by design: How design thinking transforms organizations and inspires innovation*. Harper Business.
- Carlgen, L., Rauth, I., & Elmquist, M. (2016). Framing design thinking: The concept in idea and enactment. *Creativity and Innovation Management*, 25(1), 38–57.
- Chen, T. M. (2010). Smart grids, smart cities need better networks [Editor's Note]. *IEEE Network*, 24(2), 2–3.
- City Health Dashboard. (2023). *Explore health in your city*. <https://www.cityhealthdashboard.com/>
- City of Edmonton. (n.d.). *Citizen dashboard*. <https://dashboard.edmonton.ca>
- City of Vancouver Open Data Portal. (2023). *VanDashboard*. <https://opendata.vancouver.ca/pages/vandashboard/>
- Dam, R., & Siang, T. (2018). *What is design thinking and why is it so popular?* <https://www.interaction-design.org/literature/article/what-is-design-thinking-and-why-is-it-so-popular>
- Difallah, D. E., Cudre-Mauroux, P., & McKenna, S. A. (2013). Scalable anomaly detection for smart city infrastructure networks. *IEEE Internet Computing*, 17(6), 39–47.

- Dorst, K. (2011). The core of 'design thinking' and its application. *Design Studies*, 32(6), 521–532.
- Etzkowitz, H., & Zhou, C. (2017). *The triple helix: University – industry – government innovation and entrepreneurship*. Routledge.
- Few, S. (2006). *Information dashboard design: The effective visual communication of data*. O'Reilly Media.
- Few, S. (2007). Why most dashboards fail. http://www.perceptualedge.com/brochures/Why_Most_Dashboards_Fail.pdf
- Haeusler, M. H., Asher, R., & Booth, L. (2017). Urban pinboard development of a platform to access open-source data to optimise urban planning performance. In A. Fioravanti, S. Cursi, S. Elahmar, S. Gargaro, G. Loffreda, G. Novembri, & A. Trento (Eds.) *Sharing of computable knowledge! (ShoCK!)* (1.) (pp. 439–448). eCAADe. <https://doi.org/10.52842/conf.ecaade.2017.1.439>
- Hasso Plattner Institute of Design. (2018). *Design thinking bootleg*. <https://dschool.stanford.edu/resources/design-thinking-bootleg>
- Kitchin, R., Laurialt, T. P., & McArdle, G. (2015). Knowing and governing cities through urban indicators, city benchmarking and real-time dashboards. *Regional Studies, Regional Science*, 2(1), 6–28.
- Kolko, J. (2015). Design thinking comes of age. *Harvard Business Review*, 93(9), 66–71.
- Lee, E. K., Chu, P., & Gadh, R. (2013). Fine-grained access to smart building energy resources. *IEEE Internet Computing*, 17(6), 48–56.
- Linton, G., & Kinton, M. (2019). University entrepreneurship education: A design thinking approach to learning. *Journal of Innovation and Entrepreneurship*, 8(1), 1–11.
- Lockwood, T. (Ed.) (2009). *Design thinking: Integrating innovation, customer experience, and brand value* (3rd ed.). Allworth Press.
- London Dashboard. (2023). Choose city. <https://citydashboard.org/London>
- Mattern, S. (2015). Mission control: A history of the urban dashboard. *Places Journal*. <https://placesjournal.org/article/mission-control-a-history-of-the-urban-dashboard/>
- Mattern, S. (2017). A city is not a computer. *Places Journal*. <https://placesjournal.org/article/a-city-is-not-a-computer/>
- McArdle, G., & Kitchin, R. (2016). The Dublin Dashboard: Design and development of a real-time analytical urban dashboard. *ISPRS Annals of the Photogrammetry Remote Sensing and Spatial Information Sciences*, 4, 19–25.
- Melbourne Dashboard. (2017). *Melbourne Dashboard*. <https://citydashboard.com.au/melbourne.php>
- Miller, B. H. (2017). What is design thinking? (And what are the 5 stages associated with it?) *Medium*. <https://medium.com/@bhmler0712/what-is-design-thinking-and-what-are-the-5-stages-associated-with-it-d628152cf220>
- Mineraud, J., Mazhelis, O., Su, X., & Tarkoma, S. (2016). A gap analysis of Internet-of-Things platforms. *Computer Communications*, 89, 5–16.
- Moere, A. V., & Purchase, H. (2011). On the role of design in information visualization. *Information Visualization*, 10(4), 356–371.
- Rahmat, H., Marshall, N., Steinmetz, C., Park, M., Bishop, K., Thompson, Corkery, L., & Tietz, C. (2020). The role of smart city initiatives in driving partnerships: A case study of the Smart Social Spaces project, Sydney Australia. In R. Roggema, & A. Roggema (Eds.) *Designing sustainable cities* (pp. 143–159) Springer.
- Rawsthorn, A. (2013). *Hello world: Where design meets life*. Penguin.
- Sadowski, J. (2021). Anyway, the dashboard is dead: On trying to build urban informatics. *New Media & Society*. <https://doi.org/10.1177/14614448211058455>
- Sadowski, J., & Maalsen, S. (2020). Modes of making smart cities: Or, practices of variegated smart urbanism. *Telematics and Informatics*, 55, 101449.
- Sarikaya, A., Correll, M., Bartram, L., Tory, M., & Fisher, D. (2018). What do we talk about when we talk about dashboards? *IEEE Transactions on Visualization and Computer Graphics*, 25(1), 682–692.
- Smart Dublin. (2016). *The Dublin Dashboard allows users to get detailed and updated information about the city that helps decision making and encourages evidence-informed analysis*. <https://smartdublin.ie/dublin-dashboard/>
- Sohaib, O., Solanki, H., Dhaliwa, N., Hussain, W., & Asif, W. (2019). Integrating design thinking into extreme programming. *Journal of Ambient Intelligence and Humanized Computing*, 10(6), 2485–2492.
- Steinmetz, C., Park, M., Tietz, C., Rahmat, H., Marshall, N., Thompson, S., Bishop, K., & Corkery, L. (2021). IoT integrated street furniture: A case study of university, industry and local government collaboration. *International Journal of Engineering Science Technologies*, 5(6), 43–53.
- Stimmel, C. L. (2015). *Building smart cities: Analytics, ICT, and design thinking*. Auerbach Publications.
- Tanda, A., & De Marco, A. (2018). Drivers of public demand of IoT-enabled smart city services: A regional analysis. *Journal of Urban Technology*, 25(4), 77–94.
- Tomitsch, M. (2018). *Making cities smarter: Designing interactive urban applications*. Jovis.
- University of New South Wales. (n.d.). *City analytics*. <https://cityfutures.adu.unsw.edu.au/research/city-analytics/>
- Valentine, L., Kroll, T., Bruce, F., Lim, C., & Mountain, R. (2017). Design thinking for social innovation in health care. *The Design Journal*, 20(6), 755–774.
- Xu, H., & Geng, X. (2019). People-centric service intelligence for smart cities. *Smart Cities*, 2, 135–152.
- Yigitbasioglu, O. M., & Velcu, O. (2012). A review of dashboards in performance management: Implications for design and research. *International Journal of Accounting Information Systems*, 13(1), 41–59.
- Young, G. W., & Kitchin, R. (2020). Creating design guidelines for building city dashboards from a user's perspectives. *International Journal of Human-Computer Studies*, 140, 102429. <https://doi.org/10.1016/j.ijhcs.2020.102429>
- Young, G. W., Kitchin, R., & Naji, N. (2020). Building city dashboards for different types of users. *Journal of Urban Technology*, 28(1–2), 289–301. <https://doi.org/10.1080/10630732.2020.1759994>
- Zanella, A., Bui, N., Castellani, A., Vangelista, L., & Zorzi, M. (2014). Internet of things for smart cities. *IEEE Internet of Things Journal*, 1(1), 22–32.

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